

地球物理第一次上課：中英對照開場講稿

First Class of Geophysics: Bilingual Opening Script

課程核心理念 / Core Idea

我們不是要學英文，而是用英文來學習新的學科知識；我們不只學地球物理、AI 與數位科技，也學習如何進行科學思考。

We are not learning English for its own sake. We use English to learn new scientific knowledge. We will learn not only geophysics, AI, and digital technology, but also how to think scientifically.

Geophysics + English + AI + Scientific Thinking

1. 課程開始 / Course Opening

今天是我們這學期第一次上課。

Today is our first class of this semester.

這學期，我們會同時使用中文和英文兩種語言來上課。

This semester, we will use both Chinese and English in this class.

大家不用擔心聽不懂英文。老師會盡量提供中英對照的教材與講稿，讓大家可以參考。

Please do not worry if you cannot understand everything in English. I will provide bilingual materials and scripts for you to follow.

我們這門課的目的，不是要把地球物理課變成英文課。

The purpose of this course is not to turn geophysics into an English class.

我們不是為了「學英文」而學英文，而是希望慢慢學會用英文來學習新的學科知識。

We are not learning English just for the sake of learning English. We are learning how to use English to learn new scientific knowledge.

所以，如果一開始只能聽懂一部分，沒有關係。

So, it is completely fine if you understand only part of the English at first.

重要的是，你要逐漸熟悉科學家怎麼使用英文描述現象、提出問題、表達證據，以及解釋自己的想法。

What matters is that you gradually learn how scientists use English to describe phenomena, ask questions, present evidence, and explain their ideas.

2. 我們不只是學知識，也要學習怎麼思考 / Learning How to Think

在這門課裡，我希望大家不只是記住地球物理的知識。

In this course, I hope you will do more than simply memorize geophysical knowledge.

我更希望大家學習一件重要的事情：

More importantly, I want you to learn something essential:

我們怎麼知道一件事情是對的？

How do we know whether an idea is correct?

科學不是只告訴我們答案。科學更重要的是問：你為什麼這樣認為？你的證據是什麼？你如何從證據得到這個結論？

Science is not only about giving us answers. More importantly, science asks: Why do you think so? What is your evidence? How do you connect your evidence to your conclusion?

3. 從一個地震問題開始 / Start with an Earthquake Question

例如，我現在提出一個想法：

For example, I have an idea:

我認為全世界並不是所有地方都會經常發生地震。地震通常集中發生在某些特定的地方。

I think earthquakes do not occur equally everywhere in the world. Earthquakes are usually concentrated in certain areas.

這是我的 Claim。

This is my claim.

Claim：我認為事情是這樣的。

Claim: This is what I think is happening.

4. Evidence：你的證據在哪裡？ / What Supports Your Claim?

但是，科學不能只有「我覺得」。所以接下來我要問：我的證據是什麼？

But science cannot stop at “I think so.” So the next question is: What is my evidence?

例如，我們可以觀察一張全球地震活動分布圖。

For example, we can examine a map showing the global distribution of earthquakes.

我們會發現，地震並不是平均散布在整個地球表面。很多地震集中形成一條一條的帶狀區域。

We can see that earthquakes are not evenly distributed across the Earth's surface. Many earthquakes are concentrated along narrow belts.

因此，這張全球地震分布圖支持了我剛才的說法。這就是我的 Evidence。

Therefore, the global earthquake distribution map supports my claim. This is my evidence.

Evidence：我看到了什麼資料或現象，可以支持我的說法？

Evidence: What data or observations support my claim?

5. Reasoning：為什麼會這樣？ / Why Does This Happen?

但是，我們還沒有完成。下一個問題是：為什麼地震會集中在這些地方？

But we are not finished yet. The next question is: Why are earthquakes concentrated in these areas?

我們可能會提出一個解釋：這可能與地球的板塊運動以及板塊構造有關。

We may propose an explanation: This may be related to plate motion and plate tectonics.

地球最外層並不是完整的一整塊。它可以看成由許多不同的板塊拼接而成，而這些板塊彼此之間會產生相對運動。

The outer part of the Earth is not one single unbroken shell. It can be viewed as being made of several tectonic plates, and these plates move relative to one another.

當板塊碰撞、分離或彼此錯動時，岩石會累積應力。當累積的應力突然釋放，就可能產生地震。

When plates collide, separate, or slide past each other, stress can build up in rocks. When the accumulated stress is suddenly released, an earthquake may occur.

因此，板塊運動可以解釋為什麼許多地震集中出現在特定區域。這就是我的 Reasoning。

Therefore, plate motion can help explain why many earthquakes occur in particular regions. This is my reasoning.

Reasoning：我如何利用科學原理，把 Evidence 和 Claim 連接起來？

Reasoning: How do I use scientific ideas to connect my evidence with my claim?

6. CER：一種有意識的科學思考方式 / CER as Scientific Thinking

所以剛才我們其實完成了一個非常重要的科學思考流程：Claim – Evidence – Reasoning，簡稱 CER。

So, what we have just done is an important scientific thinking process: Claim – Evidence – Reasoning, or CER.

CER	中文提問	English Prompt
Claim	我認為事情是怎樣？	What do I think is happening?
Evidence	我有哪些觀察、資料或證據？	What observations or data support my idea?
Reasoning	為什麼這些證據可以支持我的說法？	Why does this evidence support my claim?

CER 可以幫助我們有意識地進行探究與思考。它讓我們不只是記住答案，而是開始思考答案是怎麼建立出來的。

CER helps us investigate and think more consciously. Instead of simply memorizing an answer, we begin to understand how that answer is constructed.

7. 從 CER 到科學模型 / From CER to Scientific Models

當我們不斷進行 Claim、Evidence 和 Reasoning 的過程，我們其實正在逐漸形成一個對自然世界的解釋。

As we repeatedly develop claims, examine evidence, and construct reasoning, we gradually build an explanation of the natural world.

這個解釋，可以進一步形成一個科學模型。

This explanation can develop into a scientific model.

地球表面由許多板塊組成，而板塊持續運動。許多地震、火山和地形現象與板塊之間的交互作用有關。

The Earth's surface consists of several tectonic plates, and these plates are constantly moving. Many earthquakes, volcanoes, and landforms are related to interactions between these plates.

這就是一個可以幫助我們解釋現象的模型。更重要的是，一個好的模型還應該可以幫助我們進行預測。

This is a model that can help us explain natural phenomena. More importantly, a good model should also help us make predictions.

例如，如果我們知道板塊邊界的位置，我們是否可以預測哪些區域比較容易發生地震？

For example, if we know where plate boundaries are located, can we predict which regions are more likely to experience earthquakes?

8. 模型不是永遠正確的 / Models Are Not Always Perfect

模型並不是永遠正確的。有一天，我們可能發現一個新的現象，而這個現象是原本的模型沒有辦法解釋的。

Models are not always perfect. One day, we may discover a new phenomenon that our existing model cannot explain.

這時候，我們不應該假裝這個現象不存在。我們應該重新問問題、重新蒐集 Evidence、重新提出或修正 Claim、重新思考 Reasoning。

At that point, we should not pretend that the phenomenon does not exist. We should ask new questions, collect new evidence, develop or revise our claim, and reconsider our reasoning.

我們再跑一次 CER，最後可能修正原來的模型。

We go through the CER process again and may eventually revise our original model.

9. 科學建模是一個循環 / Scientific Modeling Is a Cycle

所以，科學建模不是老師告訴你一個模型，然後你把它背起來。

Scientific modeling is not simply receiving a model from the teacher and memorizing it.

觀察現象 → Claim → Evidence → Reasoning → 建立模型 → 解釋與預測 →
發現問題 → 修正模型

Observe → Claim → Evidence → Reasoning → Construct a Model → Explain and
Predict → Find Problems → Revise the Model

然後這個循環可能再重新開始。這也是科學知識不斷進步的重要方式。

Then the cycle begins again. This is one of the main ways scientific knowledge develops over time.

10. AI 在這門課裡扮演什麼角色？ / What Role Will AI Play?

這學期我們也會使用很多 AI 技術與數位科技。AI 可以幫助我們搜尋資料、整理資料、寫程式、分析資料、製作圖表、提出可能的解釋，甚至協助我們建立模型。

This semester, we will also use AI tools and digital technologies. AI can help us search for information, organize information, write code, analyze data, create figures and maps, suggest possible explanations, and even assist us in constructing models.

AI 給你答案，不代表答案就是正確的。

An answer from AI is not automatically a correct answer.

所以當 AI 提供一個答案時，我希望大家開始問：它的 Claim 是什麼？它有什麼 Evidence？它的 Reasoning 合理嗎？我可以找到其他資料來驗證嗎？

When AI gives you an answer, I want you to ask: What is its claim? What evidence does it have? Is its reasoning reasonable? Can I find other evidence to verify it?

11. AI 可以協助思考，但不能取代理解 / AI Can Support Thinking

在這門課裡，AI 是我們的協作者。它可以幫助我們工作得更快，也可以提供新的想法，但是最後做判斷的人，還是我們自己。

In this course, AI is our collaborator. It can help us work faster and provide new ideas, but we are still responsible for making the final judgment.

我們必須能夠回答：我為什麼相信這個答案？我的證據在哪裡？我能不能解釋這個結果？如果結果不合理，我要怎麼修正？

We should be able to answer: Why do I believe this answer? Where is my evidence? Can I explain this result? If the result does not make sense, how should I revise it?

真正的學習，往往就是從這些問題開始。

Real learning often begins with these questions.

12. 這學期我們真正要學的是什麼？ / What Are We Really Going to Learn?

所以這門課不只是一要教大家地球物理。我們會學習地震、地球內部構造、板塊運動，以及利用物理方法了解地球。

Therefore, this course is not only about geophysics. We will study earthquakes, the internal structure of the Earth, plate motion, and how physical methods can be used to investigate our planet.

我們也會使用 AI、程式設計、資料分析、數位工具與視覺化技術。

We will also use AI, programming, data analysis, digital tools, and visualization techniques.

但是更重要的是，我希望這門課可以教大家如何像科學家一樣思考。

But more importantly, I hope this course will teach you how to think like a scientist.

看到一個現象 → 提出問題 → 提出自己的想法 → 尋找證據 → 建立解釋 →
用資料測試 → 發現問題就修正 → 建立更好的模型

Observe a phenomenon → Ask a question → Develop an idea → Look for evidence →
Construct an explanation → Test with data → Revise when needed → Build a better
model

Closing Message / 最後一句

我們用地球物理來學習地球，用英文來接觸科學，用 AI 來協助探索，但最後用自己的思考來理解世界。

We use geophysics to study the Earth, English to access science, AI to support exploration, and our own thinking to understand the world.

Claim → Evidence → Reasoning → Model → Test → Revise